UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

# OmniFold

## Lecture 20

Course of:

**Statistics and data analysis**

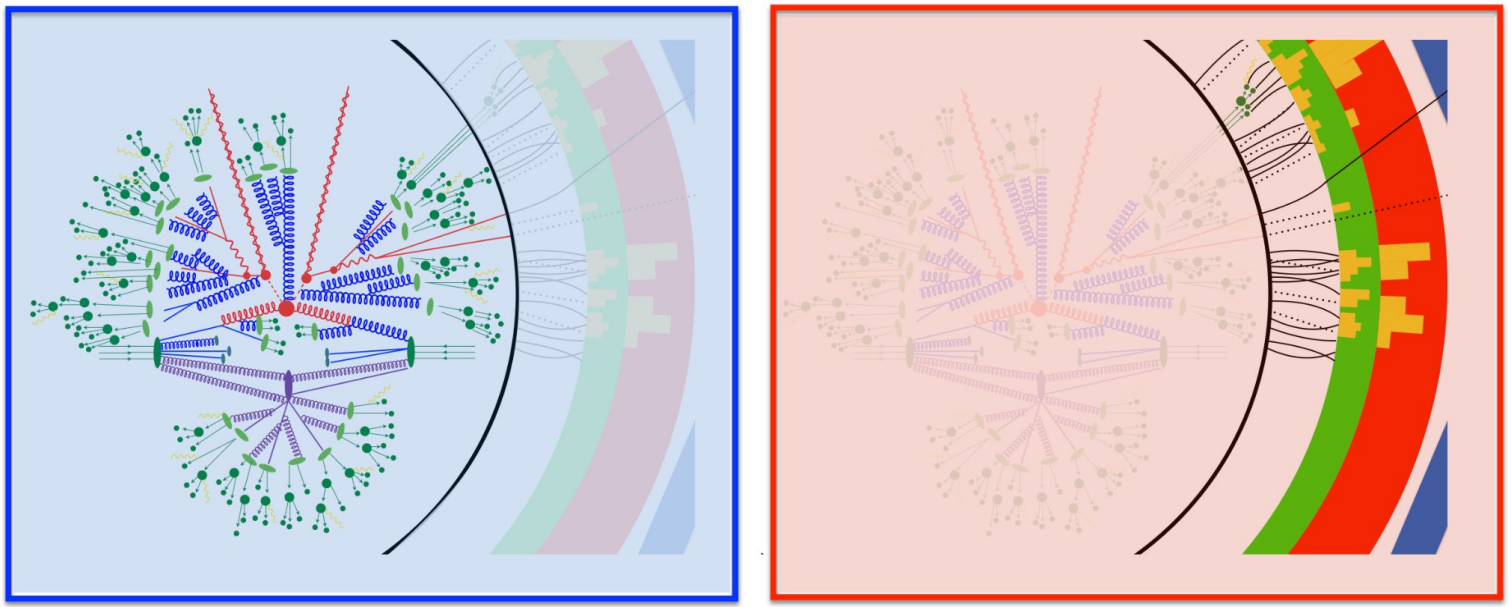
*Federico De Guio - Martino Borsato*

# Unfolding

Recap

**Want this**

**Measure this**



i.e. remove detector distortions

# Does a **Maximum Likelihood** approach work?

Recap

- If I know ***p(meas | true)*** → find the *true* that **maximise *p* given *true***
  - *p(meas | true)* is our likelihood
  - No inverse problem, this would be a forward folding approach

- When dealing with events produced at an accelerator, the typical situation is an hypervariate true space (i.e. quadrimomentum of all produced particles) and hyperspectral measured space (response from many subdetectors) → **Our likelihood is too complex!** (intractable)

- Can we bin it into a response matrix? Practical example:
  - **5 kinematic variables** associated to particles produced at an accelerator (true space)
  - **5 detector variables** such as Energy or momentum (measurement space)
  - **Only 20 bins** for each variable
  - The response matrix would explode: $20^5$ x $20^5$ ~ **$10^{13}$ bins!**
  - Hard to produce a MC able to fully populate the response matrix → **most of these bins remain empty**

- **This is what the IBU approach does** but doesn't scale because of the **curse of dimensionality**

# **Three challenges** for unfolding methods

Recap

- In classical methods, measured observables **need to be binned into histograms**
  - Binning derived ahead of time
  - Binning chosen manually

- The **response matrix often doesn't take into account all possible effects** playing a role in the detector response

- Because the measurements are binned, one can only **unfold a small number of observables simultaneously**
  - Measurements beyond two or three dimensions are simply not feasible

# **Likelihood free** approach with **reweighting**

- The idea is to work with the MC simulation, **sample single events** and build a reweighting mechanism

  - **Dataset 1**: sampled from *q(x)*
  - **Dataset 2**: sampled from *p(x)*

- Create weights ***w(x) = p(x)/q(x)*** so that when dataset 1 is weighted by w, it is statistically identical to dataset 2

- This is what I'm interested in! How can I implement it at the single event level?

# A **classification approach** for estimating the likelihood ratio

- A NN that performs binary classification is all in all a **likelihood ratio approximator** (theorem)
  - The loss function is typically the **binary cross-entropy**

$$L(f) = -\int p(x)\log(f(x))\,dx - \int q(x)\log(1-f(x))\,dx$$

*f(x)* is the NN output

  - If I minimize it I get: $f(x) = \frac{p(x)}{p(x)+q(x)}$ ⇒ $r(x) = \frac{p(x)}{q(x)} = \frac{f(x)}{1-f(x)}$

- **I train to perform classification, but I am actually building a likelihood ratio approximation**
  - No need to compute *p(x)* or *q(x)* separately
  - **Transformed a problem of density estimation** (hard) **into a problem of classification** (easy)

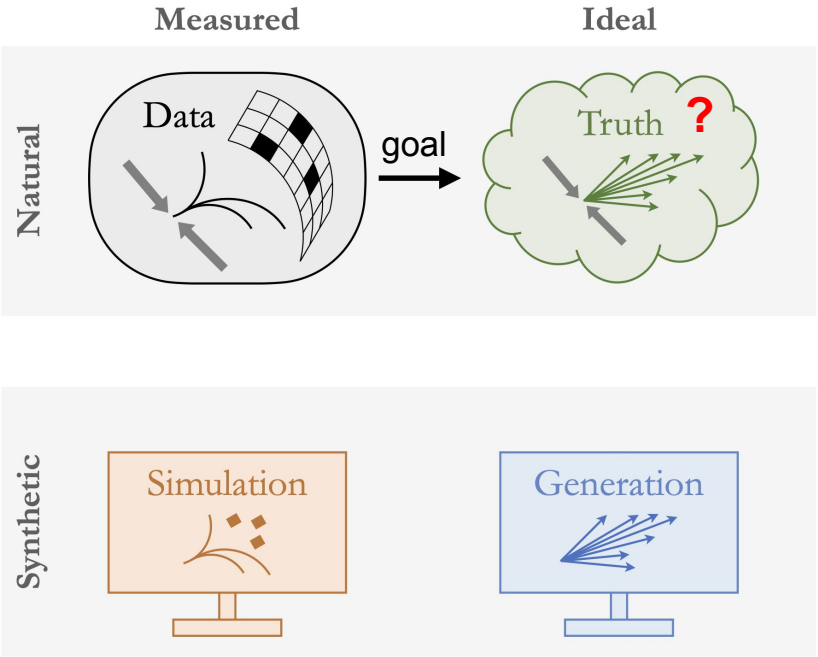
- In other words: train the NN to distinguish *p(x)* from *q(x)* and interpret the output as the likelihood ratio

# Advantages of the **event based reweighting approach**

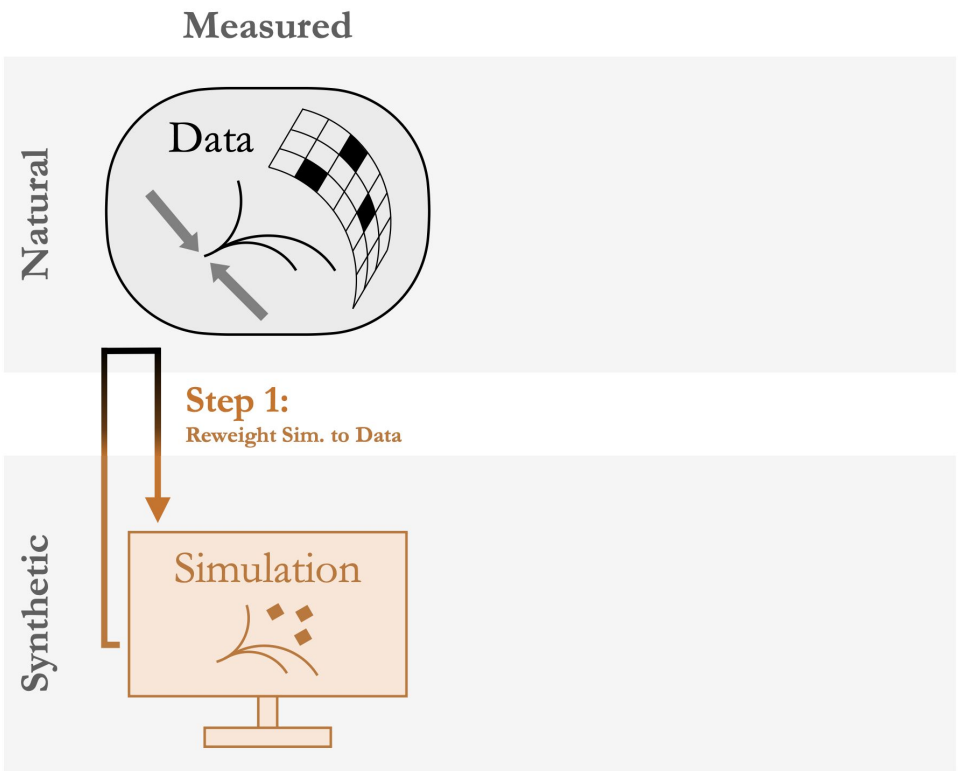
- Once I can access the **Likelihood ratio *r(x)* I can use it to reweight dataset 1** on an event-by-event basis to resemble dataset 2

- Use multiple observable in my classification
  - **Access to full phase space**
  - **No need to unfold one variable at the time**

- Thinking at our unfolding problem, we need:
  - A simulation of the physics phenomenon
  - A simulation of the detector response

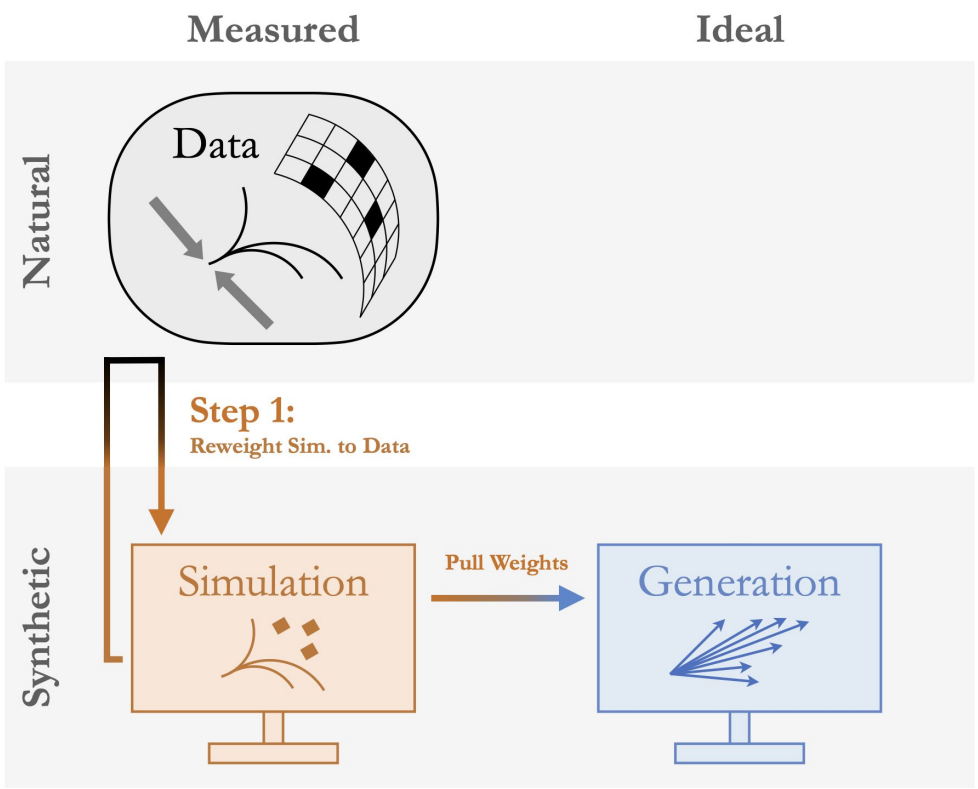


# **OmniFold**: unfold by iterating



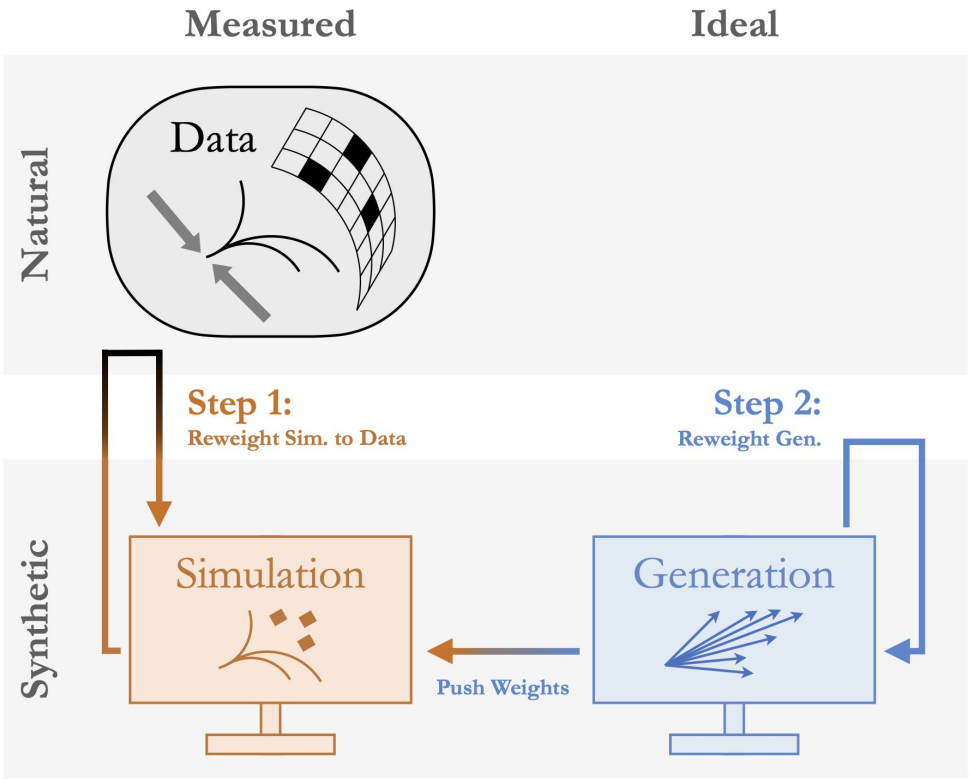

- Reweight the detector simulation to the measured data
  - Get and save the save of weights

# **OmniFold**: unfold by iterating



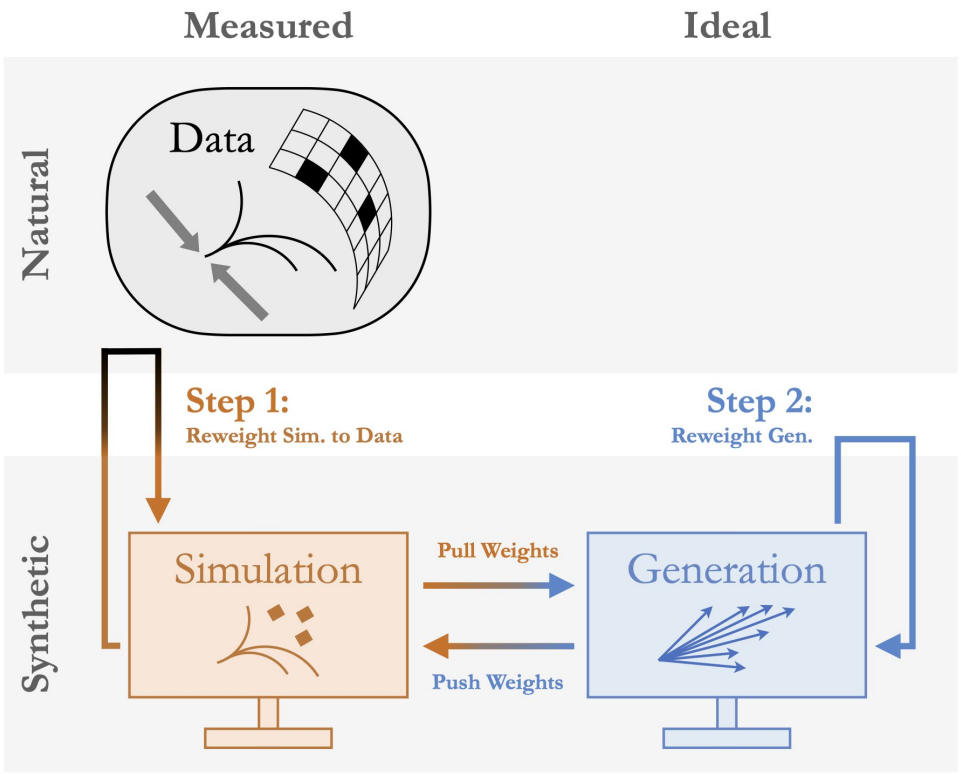

- Reweight the detector simulation to the measured data
  - Get and save the save of weights
- Pull the weights back to the pre-detector space
  - The weights are extracted in the detector space, so they don't belong here

# **OmniFold**: unfold by iterating



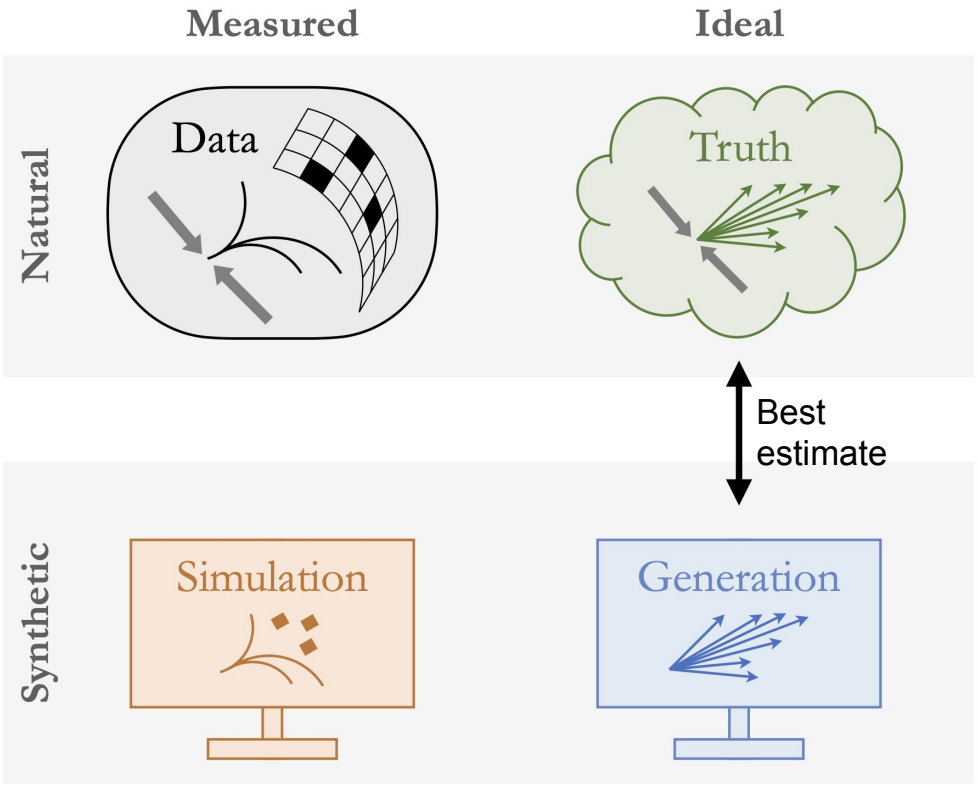

- Reweight the detector simulation to the measured data
  - Get and save the save of weights
- Pull the weights back to the pre-detector space
  - The weights are extracted in the detector space, so they don't belong here
- Train a second classifier to extract new residual weights from the classification of pre-detector events and re-weighted pre-detector events
  - This is performing smoothing

# **OmniFold**: unfold by iterating



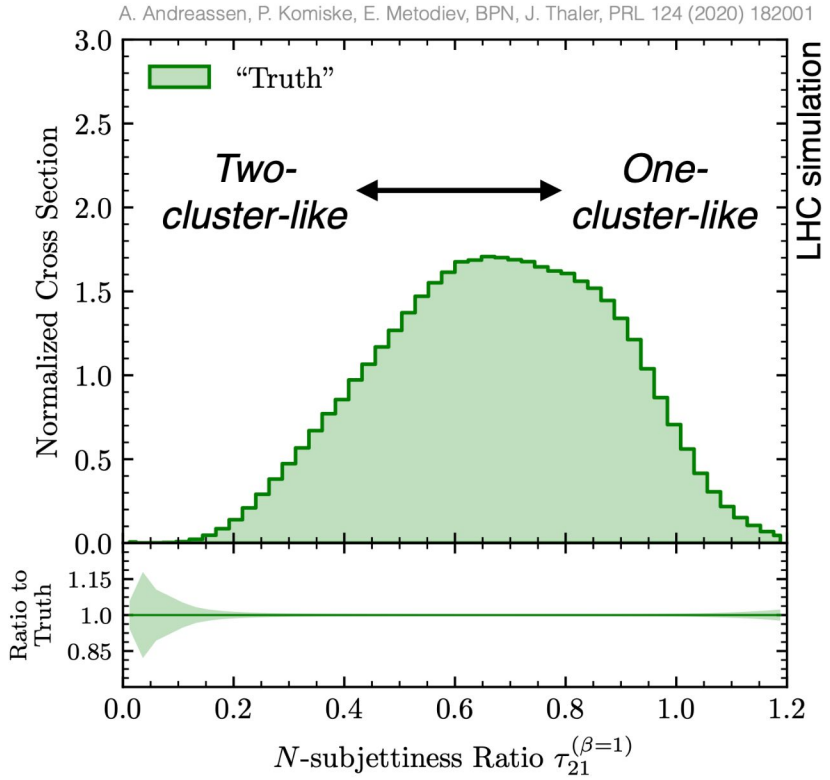

- Reweight the detector simulation to the measured data
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- Repeat

# **OmniFold**: unfold by iterating



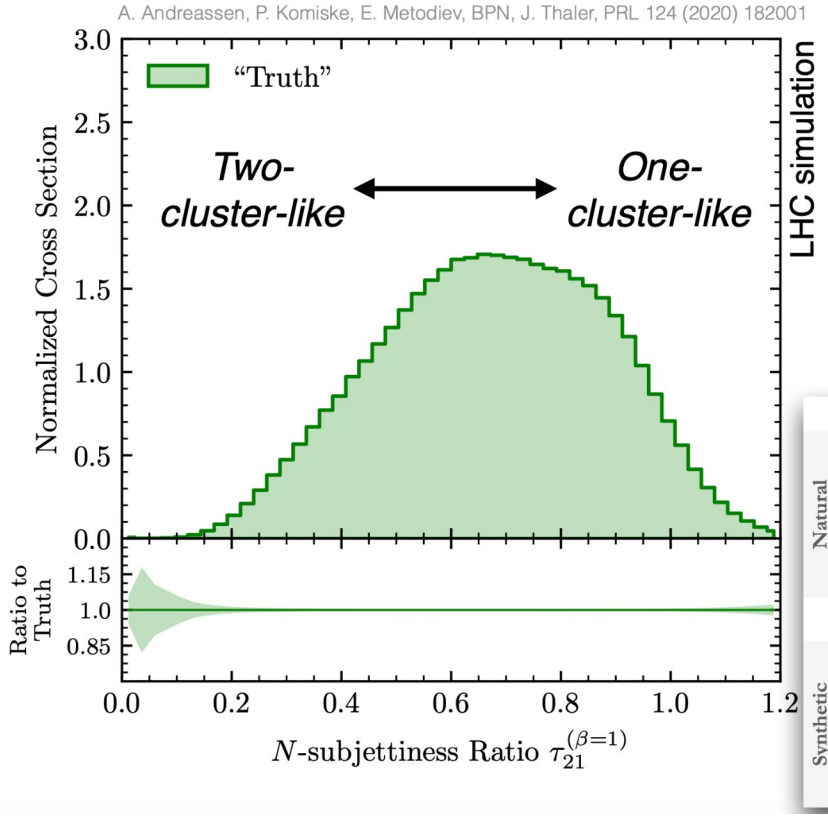

- Reweight the detector simulation to the measured data
  - Get and save the save of weights
- Pull the weights back to the pre-detector space
  - The weights are extracted in the detector space, so they don't belong here
- Train a second classifier to extract new residual weights from the classification of pre-detector events and re-weighted pre-detector events
- Repeat and stop. The weighted generator events are our best estimate

# OmniFold at work

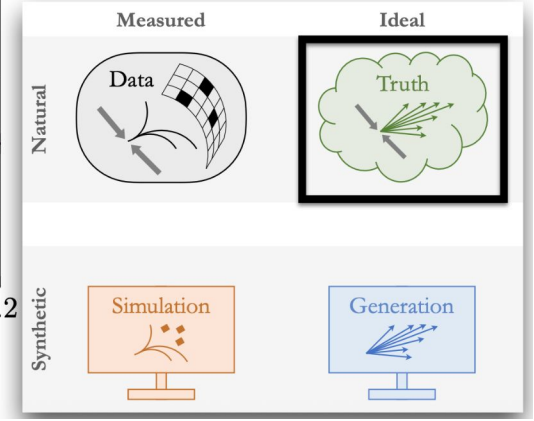



Consider this observable, which characterizes the substructure

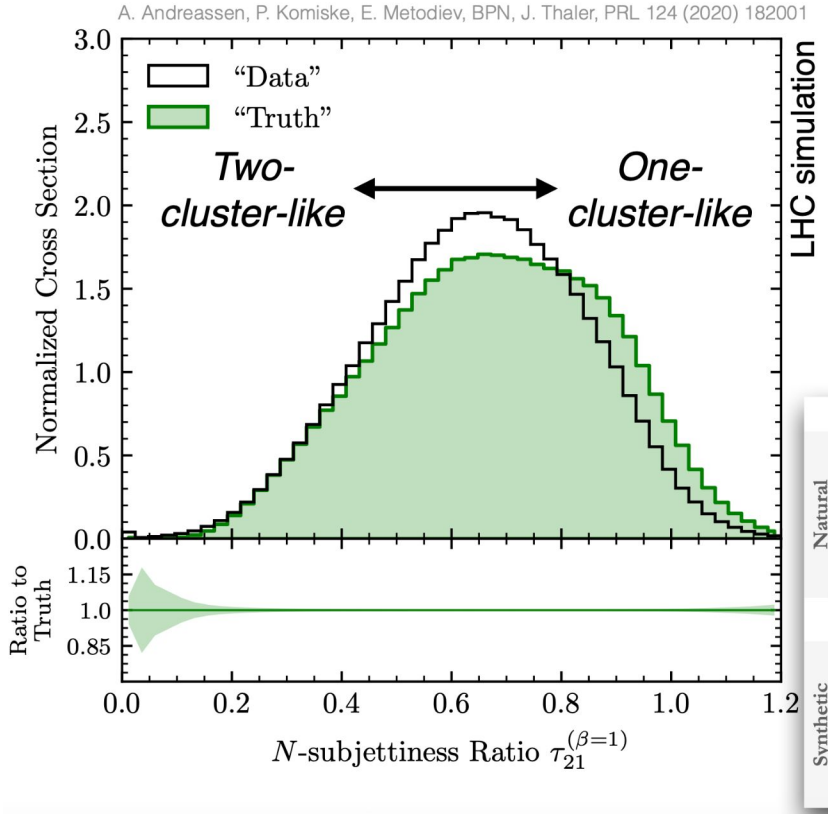
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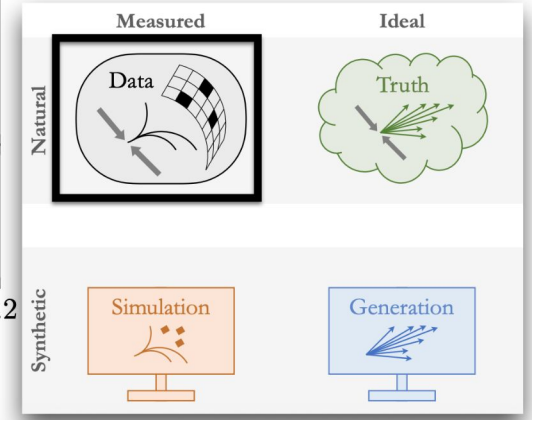

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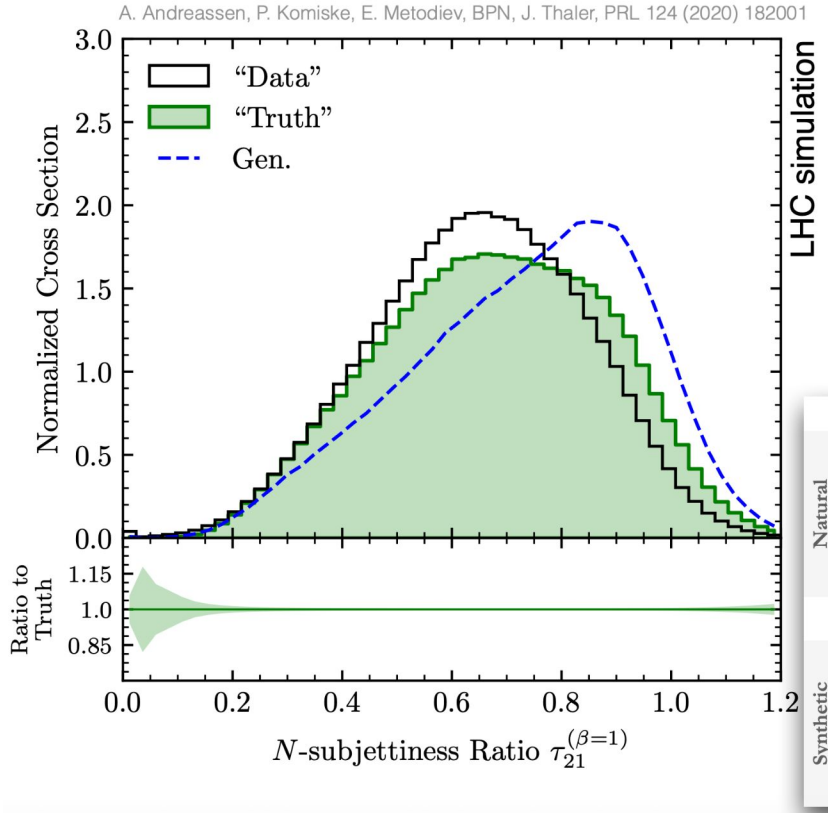
# OmniFold at work



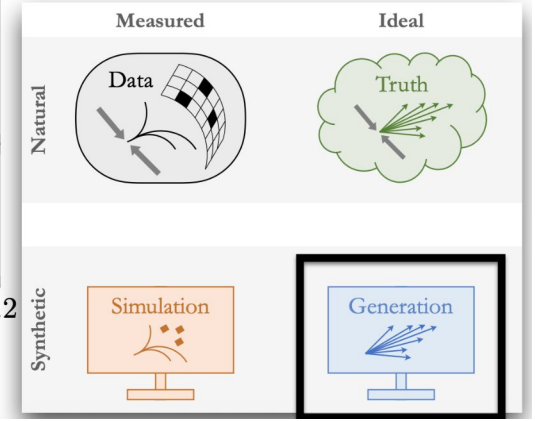

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# OmniFold at work




Consider this observable, which characterizes the substructure

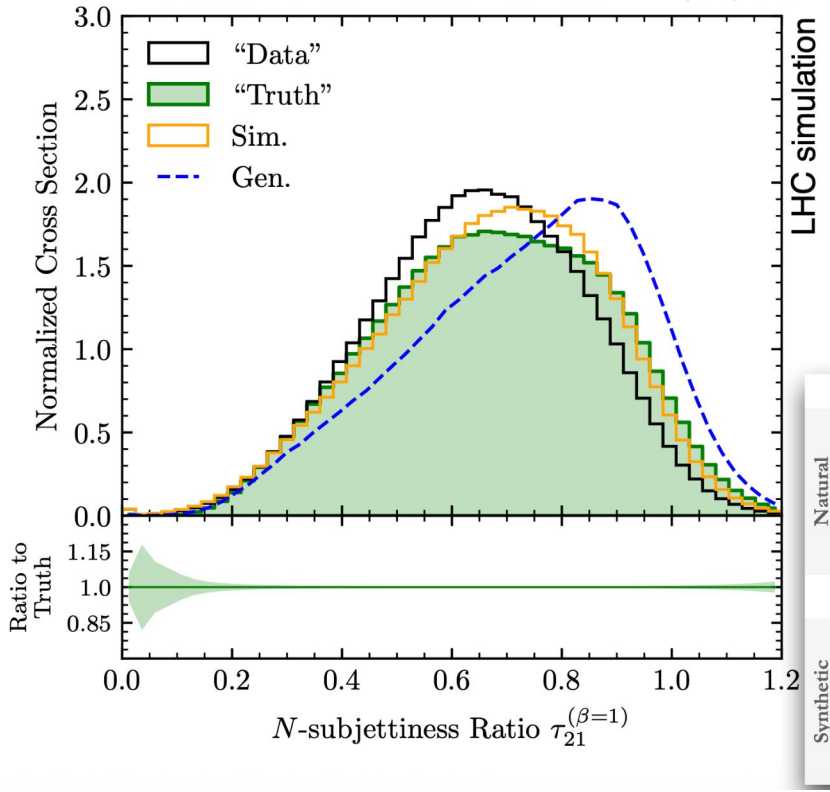


# OmniFold at work

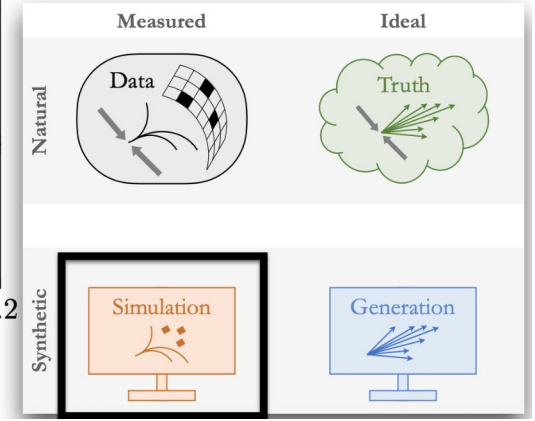
Use the mapping between blue and orange to infer the mapping between black and green

Assumption: the distorsions have to be modeled well

A. Andreassen, P. Komiske, E. Metodiev, BPN, J. Thaler, PRL 124 (2020) 182001

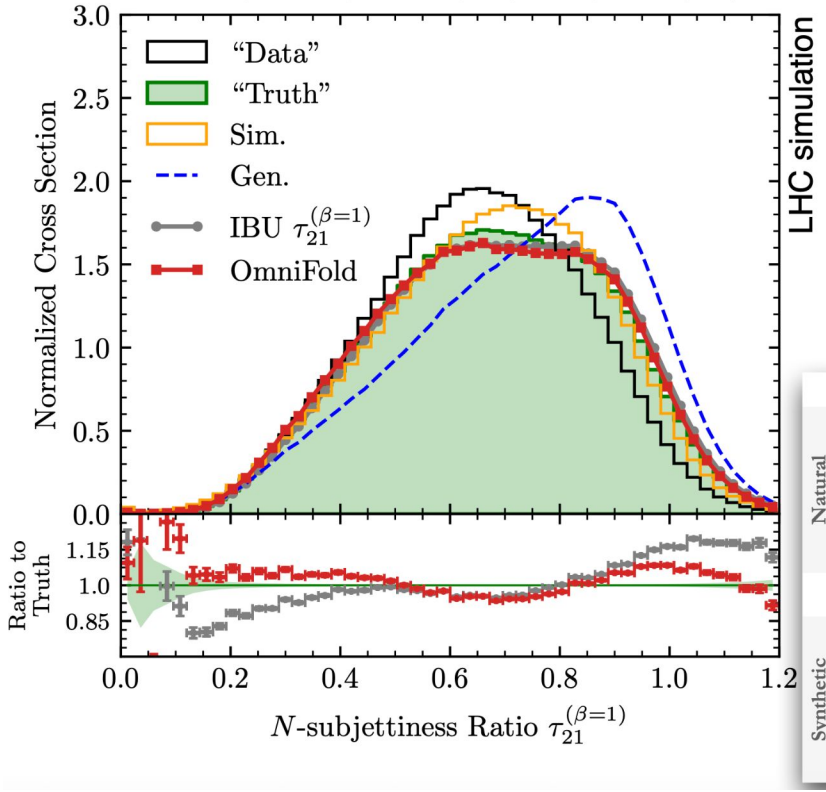



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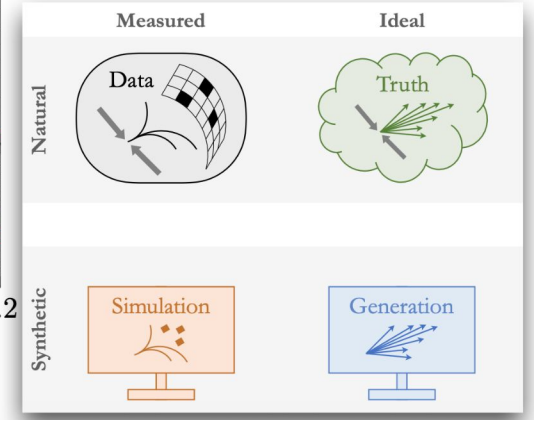


# OmniFold at work

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OmniFold outperforms IBU even though it is not tailored to this observable



# In summary

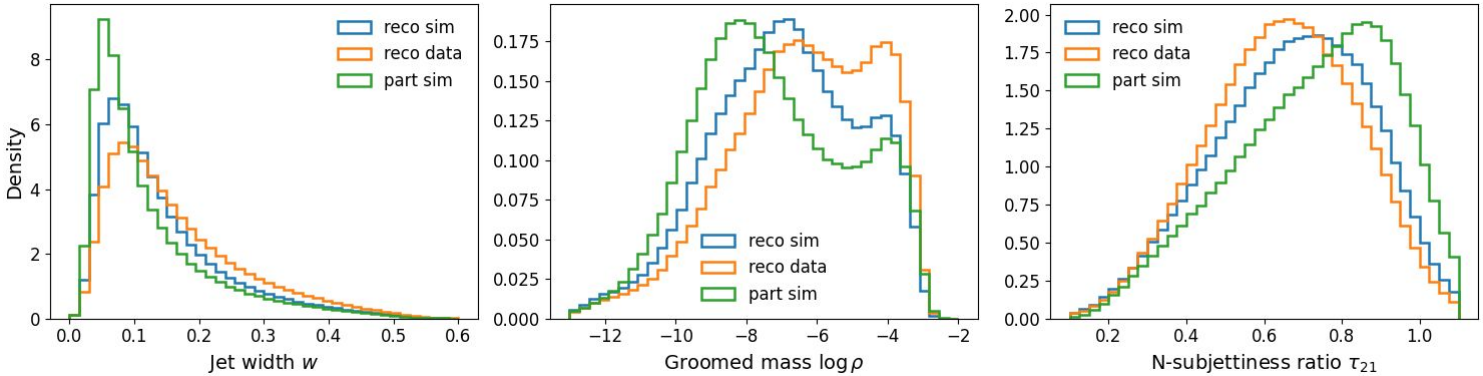
- Full phase space access → exploits correlations
- Likelihood free
- Unbinned
- Event based
- Fast

- **In a sense OmniFold is the unbinned multi-dimensional and NN powered version of the IBU**
  - They both give up on reversing the problem and adopt a forward folding approach

# Exercises…

# Implement OmniFold and unfold three observables simultaneously

- A dataset is provided where three jet-related variables are available
  - Part sim → MC events before convolving the detector response (Generation level)
  - Reco sim → same MC events after convolving the detector response
  - Reco data → the actual data measured by the detector
  - The true gen level information is also provided for comparison




- A guideline notebook is available here